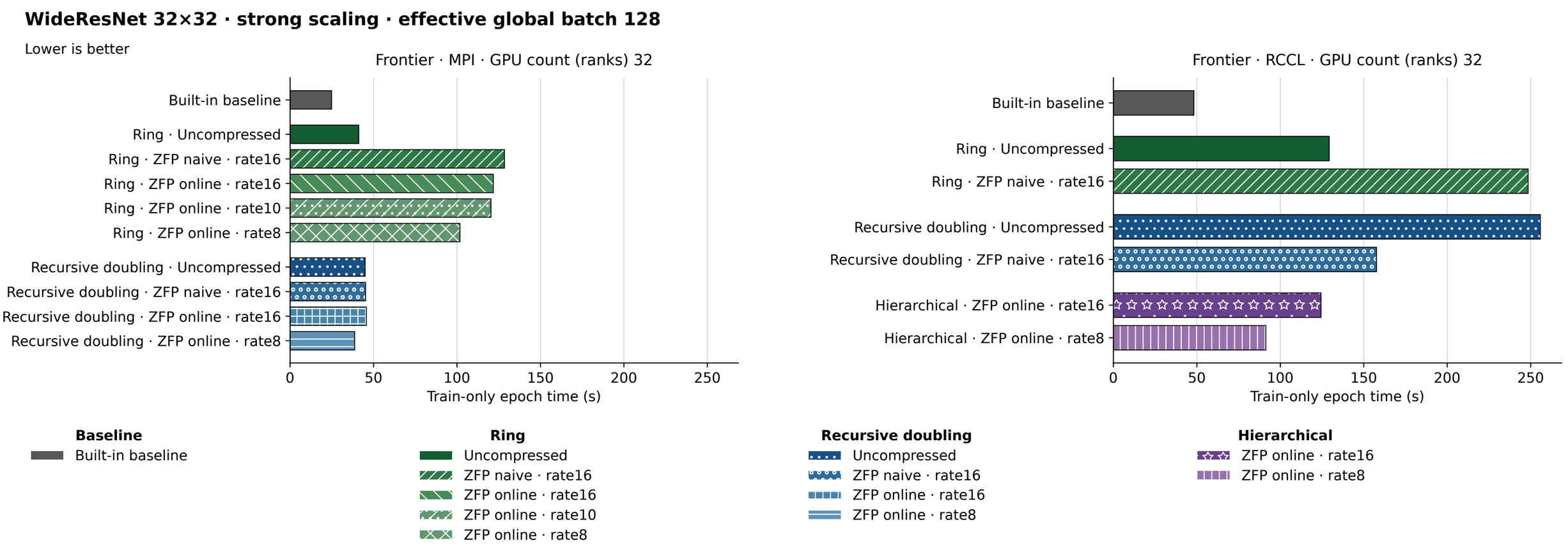

WideResNet 32×32 · strong scaling · effective global batch 128
Lower is better
Frontier · MPI · GPU count (ranks) 32
Built-in baseline
Ring · Uncompressed
Ring · ZFP naive · rate16
Ring · ZFP online · rate16
Ring · ZFP online · rate10
Ring · ZFP online · rate8
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Recursive doubling · ZFP online · rate16
Recursive doubling · ZFP online · rate8
0
50
100
150
200
250
Train-only epoch time (s)
Frontier · RCCL · GPU count (ranks) 32
Built-in baseline
Ring · Uncompressed
Ring · ZFP naive · rate16
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Hierarchical · ZFP online · rate16
Hierarchical · ZFP online · rate8
0
50
100
150
200
250
Train-only epoch time (s)
Baseline
Built-in baseline
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate